

⊕ Write the density function of the normal curve :-

⇒ The density function of the normal curve, also known as the gaussian distribution, is known by the formula:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} * e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

Here,

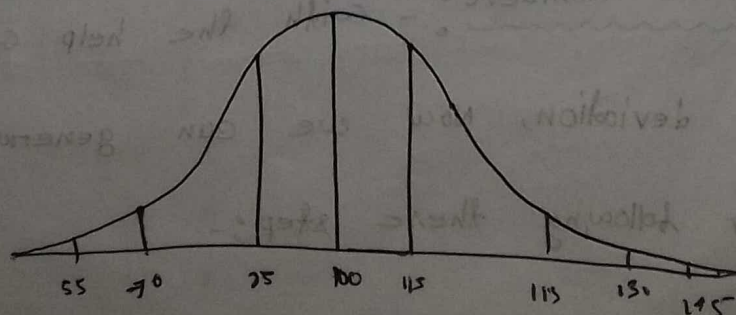
$f(x)$ = represent the density of normal distribution

μ = is the mean or average distribution

σ = is represent standard deviation, which measures the spread of the distribution.

π = π represent the value 3.1416

The formula described that, a bell-shaped curve centered around the mean (μ). The standard deviation σ determines the width of the curve.



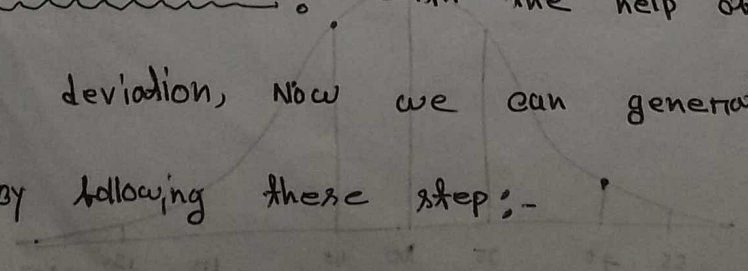
Explain the process of random number generation of the random observation from Normal distribution with a mean 15 and standard deviation 5.

⇒ Before generating random numbers, we have to understand what is normal distribution, mean and standard deviation.

Normal Distribution:- The normal distribution is a probability distribution that is shaped like a bell curve. It is characterized by its mean and standard deviation.

Mean & standard deviation:- Mean represents the average or typical value and standard deviation represents how much the values will vary around mean.

Generate random numbers:- With the help of mean and standard deviation, now we can generate random numbers:- by following these steps:-



⊕ multiple each random number by the standard deviation in this case : 5.

⊕ Then add the mean value , in this case it is 15.

⊕ Now Repeat the process for each observation .

Standard Deviation = 5

Mean = 15

random Number = standard Deviation * num + mean

⊕ Repeat the process

for example:

18.2

16.7

13.1

19.9

12.5

16.3

14.9

14.4

17.2

11.2

⊕ what is stochastic process ?

Deterministic process :- The process in which the outcome of an activity can be described completely in terms of its input is deterministic and the activity is said to be deterministic activity.

Example: In case of an automatic machine, the output per hour is a deterministic process.

Stochastic process : when the outcome of an activity is random, that is there can be various possible outcomes, then the activity is said to be stochastic.

Example: In a repair shop, the number of machines repaired will vary from hour to hour.

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poisson distribution

⊕ write the probability mass function of poisson distribution:

⇒ poisson distribution is the discrete version of Exponential distribution. Let x be a random variable which takes non-negative integer values only. Then the probability mass function

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

is called the poisson distribution with parameter λ , $\lambda > 0$

The properties of poisson distribution are:

where,

$P(x)$ = Defines the probability of observing x events

$e = e$ is the mathematical term equal to ≈ 2.718

λ = is the average rate of events

x = is the number of events

The probability mass function (PMF) of a poisson distribution gives the probability of obtaining a specific number of events in a fixed interval of time.

List several application of poisson distribution:-

1. Modeling rare event:- poisson distribution is used to model rare events that occur randomly over time. For example:- modeling the number of accidents, natural disasters.

3. Queueing Theory:- it helps to model the arrival of customers at a service facility.

it helps to analyze waiting time, resource management.

4. Call center analysis:

5. Network traffic model:- To model arrival and departure of network packet.

6. Radio active decay:-

Example 5 :- A pair of fair dice is rolled once :-

The sum of the two numbers on the dice x will range from

$$1+1=2 \text{ to } 6+6=12$$

<u>x</u>	<u>ways</u>	<u>Example</u>
2	1	(1+1)
3	2	(2+1) (1+2)
4	3	(3+1) (2+2) (1+3)
5	4	(4+1) (3+2) (2+3) (1+4)
6	5	(5+1) (4+2) (3+3) (2+4) (1+5)
7	6	(6+1) (5+2) (4+3) (3+4) (2+5) (1+6)
8	5	(5+3) (4+4) (3+5) (2+6) (6+2)
9	4	(4+5) (6+3) (5+4) (3+6)
10	3	(6+4) (4+6) (5+5)
11	2	(5+6) (6+5)
12	1	(6+6)

Total ways = 36

x	$P(x)$
2	$\frac{1}{36}$
3	$\frac{2}{36}$
4	$\frac{3}{36}$
5	$\frac{4}{36}$
6	$\frac{5}{36}$
7	$\frac{6}{36}$
8	$\frac{5}{36}$
9	$\frac{4}{36}$
10	$\frac{3}{36}$
11	$\frac{2}{36}$
12	$\frac{1}{36}$

probability that x is an odd number:

$$\Rightarrow \frac{2}{36} + \frac{4}{36} + \frac{6}{36} + \frac{4}{36} + \frac{2}{36} = \frac{18}{36} = \frac{1}{2} \text{ or } 50\%.$$

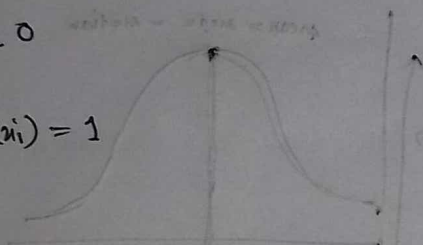
⊕ probability or mass function Discrete probability function:

If a random variable x can take $x_i (i=1, n)$, then the probability of value x_i , $p(x_i)$ is said to be the Discrete probability function or probability or mass function.

$p(x_i)$ must satisfy two problem:

(i) $p(x_i) \geq 0$

(ii) $\sum_{i=1}^n p(x_i) = 1$



probability Density Function ⊕ Continuous probability function

If the random variable is continuous and not limited to discrete values, it will have infinite number of values. It such variable is

defined by a function $f(x)$. Then

This function is called probability density function (PDF)

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

Measures of probability function

Two important characteristics of a probability or density function (PDF) are:

(1) Central Tendency

⊕ Mean

⊕ Mode

⊕ Median

(2) Dispersion

⊕ Mean Deviation

⊕ Standard Deviation

⊕ Variance

Central Tendency

Mean :-

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

If the N observation are divided into I group

$$\bar{x} = \frac{1}{N} \sum_{i=1}^I n_i x_i$$

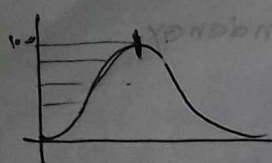
⊕ For discrete variable its probability is given

$$\bar{x} = \sum p(x_i) x_i$$

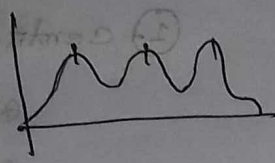
⊕ For continuous variable its probability is given

$$\bar{x} = \int_{-\infty}^{\infty} f(x) x dx$$

Mode :- The highest peak, most frequently occurring value is called the mode of a distribution.



unimodal distrib



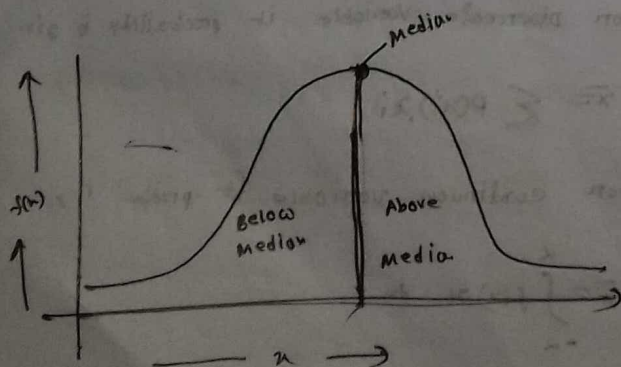
multimodal

mode is determined as:

$$\frac{d}{dx} [f(x)] = 0$$

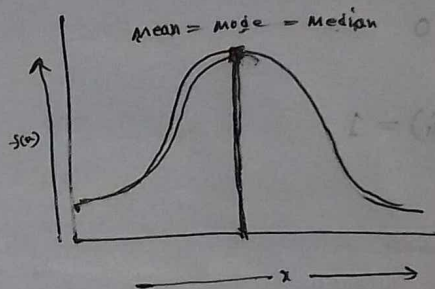
$$\frac{d^2}{dx^2} [f(x)] = 0$$

Median :- median divides the observations of the variable into two equal parts. Half value of random variable will fall below median and half above the median.

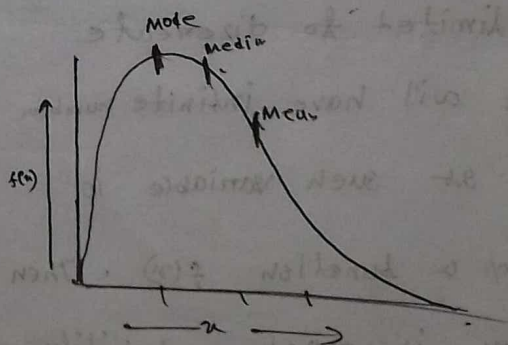


⊕ Mean, mode, median depend upon the shape of the probability density function:

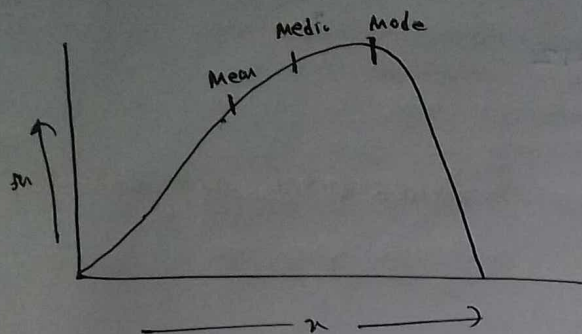
⇒ If the curve is symmetric about the center and is unimodal. Then, all three mean, mode, median will coincide.



⇒ If the curve is skewed to the left, mean is greater than mode.



⊕ If the curve skewed to the right
mode is greater than mean.

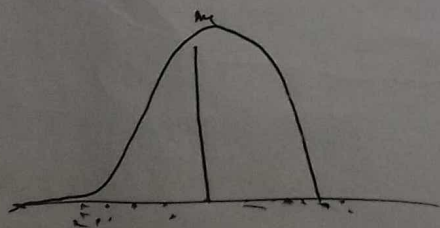


Dispersion

~~1. Mean deviation~~

Dispersion indicates the spread of the distribution.

1. Mean deviation
2. Standard deviation
3. Variance



Uniform Distribution :-

The uniform probability law states that all the values in a given interval are equally likely to occur.

suppose service times are uniformly distributed between 30 and 80 sec.

It means that a service will not take less than 30 seconds and more than 80 seconds. On the average

time will be 55 sec.

$$x = a + (b-a)r$$

where r is a random number.

Example: 5.9 → Generate three random variates from a normal distribution with mean 20 and standard deviation 5. Take $n=12$ for each observation. Take $n=12$



⇒

$$Y = \mu + d \left(\sum_{i=1}^n r_i - \frac{n}{2} \right)$$

$$Y = \mu + d \left(\sum_{i=1}^{12} r_i - 6 \right) \text{ when } n=12$$

where μ is mean = 20

d is S.D = 5

Since $n=12$, we need 12 random numbers for generating one value of the variable.

① Twelve random numbers are taken from a random number

.483, .517, .063, .229, .807, .562, .066, .924, .511, .134, .657

$$\sum_{i=1}^{12} r_i = 5.553$$

$$\therefore Y_1 = 20 + 5 (5.553 - 6) \\ = 17.775$$

② 12 random numbers are taken

.357, .944, .733, .345, .063, .546, .127, .925, .247, .816, .761

$$\sum_{i=1}^{12} r_i = 6.294$$

$$Y_2 = 20 + 5 (6.294 - 6) = 21.47$$



* Joint probability mass function :

$$P(x, y) = P(X=x, Y=y) \text{ for all } x, y$$

Discrete random variable

$\Rightarrow X, Y$ are independent if

$$P(x, y) = P_X(x) P_Y(y)$$

Example 4.9 :- suppose X and Y are

Jointly discrete random variable

$$\text{with } P(x, y) = \begin{cases} \frac{xy}{27} & \text{for } x=1, 2 \\ & y=2, 3, 4 \end{cases}$$

Then,

$$P_X(x) = \sum_{y=2}^4 \frac{xy}{27}$$

$$= \frac{2x}{27} + \frac{3x}{27} + \frac{4x}{27}$$

$$= \frac{9x}{27}$$

$$= \frac{x}{3}$$

Now,

$$P_Y(y) = \sum_{x=1}^2 \frac{xy}{27}$$

$$= \frac{y}{27} + \frac{2y}{27}$$

$$= \frac{3y}{27} = \frac{y}{9}$$

since,

$$P(x, y) = \frac{xy}{27} = P_X(x) \cdot P_Y(y)$$

$$= \frac{x}{3} \times \frac{y}{9}$$

$$= \frac{xy}{27}$$

The random variable X and Y are independent.

Joint probability density function

$f(x, y)$ of X and Y .

which determines if X and Y are jointly continuous for all sets of real number A and B .

$$P(X \in A, Y \in B) = \int_B \int_A f(x, y) dx dy$$

In this case,

X and Y are independent if

$$f(x, y) = f_X(x) \cdot f_Y(y)$$

when,

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Example: 4.11 :-

$$f(x, y) = \begin{cases} 24xy & \text{for } x \geq 0 \\ 0 & \text{for } y \geq 0 \\ x+y \leq 1 \end{cases}$$

Then

$$f_X(x) = \int_0^{1-x} 24xy \, dy$$

$$= \left[\frac{24xy^2}{2} \right]_0^{1-x}$$

$$= \left[12xy^2 \right]_0^{1-x}$$

$$= 12x(1-x)^2$$

$$= 12(1-x)^2$$

$$f_Y(y) = \int_0^{1-y} 24xy \, dx$$

$$= \left[12x^2y \right]_0^{1-y}$$

$$= 12y(1-y)^2$$

$$\text{Now } f(x, y) = f_X(x) \cdot f_Y(y)$$

2

X and Y are not independent.

Chapter: 6

⊕ Uniform/Rectangular Distribution :

* દેખાતે જાણી particular interval
જે મલકે દેખાતે જાણી function જે
value મળે k થાય, તેથી uniform
distribution.

⊕ probability Density Function (PDF) :

$$f(x) = \begin{cases} k & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

⊕ pdf property :

$$\int_{-a}^a f(x) dx = 1$$

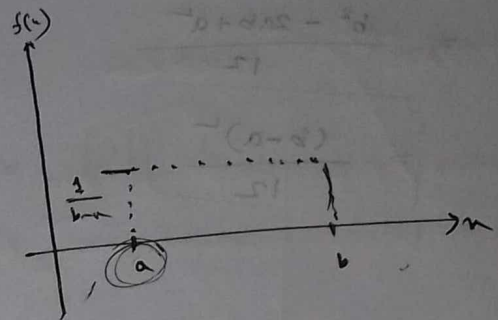
$$\Rightarrow \int_a^b k dx = 1$$

$$\Rightarrow k [x]_a^b = 1$$

$$\Rightarrow k(b-a) = 1$$

$$\Rightarrow k = \frac{1}{b-a}$$

$$\therefore f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$



Mean

$$E(x) = \int_a^b x f(x) dx$$

$$E(x) = \int_a^b x \cdot \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b$$

$$= \frac{1}{b-a} \cdot \left[\frac{b^2 - a^2}{2} \right]$$

$$= \frac{b+a}{2}$$

$$\text{pdf} \quad f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \end{cases}$$

⊕ Variance :-

$$\text{Var}(u) = E(x^2) - (E(u))^2$$

$$= \frac{b^2 + ab + a^2}{3} - \frac{(a+b)^2}{4}$$

$$= \frac{b^2 - 2ab + a^2}{12}$$

$$= \frac{(b-a)^2}{12}$$

$$E(u) = \frac{1}{b-a}$$

$$E(x^2) = \int_a^b x^2 f(u) du$$

$$= \int_a^b x^2 \cdot \frac{1}{b-a} du$$

$$= \frac{1}{b-a} \cdot \left[\frac{x^3}{3} \right]_a^b$$

$$= \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3}$$

$$= \frac{1}{b-a} \cdot \frac{(b-a)(b^2 + ab + a^2)}{3}$$

$$= \frac{b^2 + ab + a^2}{3}$$

character

Application

Density

Distribution

parameter

Range

Mean

Variance

Mod

characteristics of uniform distribution :-

Application $\rightarrow$ Essential in generating random values from all other distribution.

Density $\rightarrow f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$

Distribution $\rightarrow F(x) = \begin{cases} \frac{x-a}{b-a} & a \leq x \leq b \\ 0 & x < a \\ 1 & b < x \end{cases}$

CDF

parameter $\rightarrow$ $a \rightarrow$ Location parameter
 $b-a \rightarrow$ is scale parameter

Range $\rightarrow (a, b)$

Mean $\rightarrow \frac{b+a}{2}$

Variance $\rightarrow \frac{(b-a)^2}{12}$

Mode $\rightarrow$ Does not uniquely exist.

Exponential Distribution

A continuous random variable x

Variance :

which has the following p.d.f

$$f(x) = \lambda e^{-\lambda x} \quad 0 < x < \infty$$

is called Exponential variable

its distribution is called expo dist

$$\text{Var}(x) = E(x^2) - (E(x))^2$$

$$= \frac{2}{\lambda^2} - \frac{1}{\lambda^2}$$

$$= \frac{1}{\lambda^2}$$

$$= \sigma^2$$

Mean :

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^{\infty} x^2 e^{-\lambda x} dx$$

$$= \lambda \frac{2}{\lambda^2}$$

$$= \frac{2!}{\lambda}$$

$$= \frac{1}{\lambda}$$

$$= \mu$$

$$\int_0^{\infty} x^{n-1} e^{-ax} dx = \frac{\Gamma(n)}{a^n}$$

$$n-1+1$$

$$\frac{n!}{a^n}$$

$$\frac{(n-1)!}{a^{n-1}}$$

Characteristics of Exponential Distribution :-

- # Applications $\rightarrow$ ① Interarrival times of customers to a system.
② Time to failure of a piece of equipment.

Density $\rightarrow f(x) = \begin{cases} \frac{1}{\beta} e^{-\frac{x}{\beta}} & x \geq 0 \\ 0 & x < 0 \end{cases}$

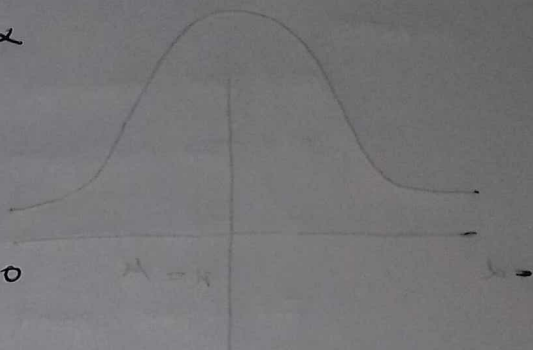
parameter $\rightarrow$ scale parameter $\beta > 0$

Range $\rightarrow [0, \infty]$

Mean $\rightarrow \frac{1}{\lambda} = \beta$

Variance $\rightarrow \frac{1}{\lambda^2} = \beta^2$

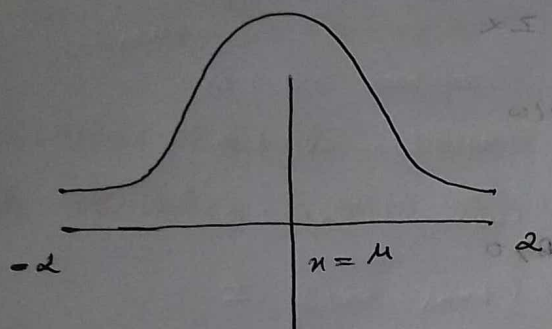
Mode $\rightarrow 0$



Normal Distribution

$$\text{Pdf, } f(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

where $-\infty < x < \infty$



Mean

$$E(x) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$= \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} dx$$

A continuous random variable which has the following p.d.f

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{1}{2} \cdot \left(\frac{x-\mu}{\sigma} \right)^2}$$

is called normal var & its distribution is called N. dis.

characteristics of Normal distribution

Density function $\rightarrow f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Distribution $\rightarrow$ No closed form

parameters $\rightarrow$ Location parameter
Scale parameter

$\mu \in (-\infty, \infty)$

$\sigma > 0$

Range $\rightarrow (-\infty, \infty)$

mean

μ

variance

σ^2

mode

μ

Chapter: 7

Linear Congruential Generator (LCG) :-

$$z_i = (a z_{i-1} + c) \bmod m$$

where

m (The modulus)

a (The multiplier)

c (The increment)

z_0 (The seed)

$$x_i = (a x_{i-1} + b) \bmod m$$

$$a \equiv b \pmod{n}$$

$$12 \equiv 0 \pmod{4}$$

a, b

$m -$

x_0

$$m = 16, a = 5, c = 3 \text{ and } z_0 = 7$$

i	$z_i = (a z_{i-1} + c) \bmod m$	$k \geq z_i / m$
0	$5 \times 7 + 3$ 7	
1	$z_1 = (5 \cdot 7 + 3) \bmod 16$	
2	$\Rightarrow \frac{39}{16} = 2$	
3	$z_2 = (5 \cdot 6 + 3) \bmod 16$	
4	$\Rightarrow \frac{33}{16} = 2$	